

Finite Element Methods – APE1

Ritz Method for a Clamped Plate

1 Theory of thin Kirchhoff plates – pure bending and transversal loads

A thin plate is a flat structure whose thickness is much smaller than its in-plane dimensions (for example metal sheets, glass panels or floors). When subjected to a transverse load, the plate bends. The unknown of the problem is the vertical deflection

$$u(x, y),$$

which measures the displacement of the mid-surface of the plate.

In the *Kirchhoff–Love plate theory*, the elastic energy depends on the *curvature* of the plate rather than on its slope. The mechanical behavior is governed by the bending rigidity

$$D = \frac{Eh^3}{12(1 - \nu^2)},$$

where

- E is the Young modulus,
- ν is the Poisson ratio,
- h is the thickness of the plate.

The equilibrium equation reads

$$D\nabla^4 u = f,$$

where f is the transverse load (force per unit area).

The bending stiffness D is supposed to be constant.

In this practical session we consider a *clamped plate*, meaning that the boundary is fixed both in displacement and rotation:

$$u = 0 \quad \text{and} \quad \partial_n u = 0 \quad \text{on } \partial\Omega.$$

2 Ritz for thin Kirchhoff plate

The goal of the exercise is to approximate the plate deflection using the Ritz method by constructing admissible trial functions that automatically satisfy these boundary conditions.

1. Write a weak/variational form of the Kirchhoff plate model. Natural boundary conditions should appear "naturally".
2. Write the Kirchhoff plate model as a minimization of a functional $\Pi(u)$.

3. Without yet having studied the theory of functional spaces, try to infer the form of the functional space in which the admissible functions for Kirchhoff plates live.
4. Find a polynomial function $w(x)$ such that

$$w(0) = 0, \quad w(L) = 0, \quad w'(0) = 0, \quad w'(L) = 0.$$

5. Using $w(x)$, construct an admissible function $\phi_1(x, y)$ that satisfies essential boundary conditions.
6. Using $U_h = \{\phi_1\}$ as a function space of size 1, use the Ritz method to approximate $u(x, y)$ – minimize $\Pi(u_h)$, $u_h \in U_h$. Feel free to use a symbolic system (wolfram, sympy) or an AI for help; you will never be asked to do this “by hand” in an exam. However, you must still be able to write the entries of the stiffness matrix K_{ij} as definite integrals.
7. Consider

$$\phi_2(x, y) = \phi_1(x, y) (x(L - x) + y(L - y)).$$

Show that ϕ_2 is admissible i.e. it satisfies the clamped boundary conditions.

8. Explain why ϕ_1 and ϕ_2 are linearly independent.
9. Using $U_h = \{\phi_1, \phi_2\}$ as a function space of size 2, use Ritz method to approximate $u(x, y)$.
10. Can you find the analytical solution of the problem using separation of variables? Is the analytical solution easy to compute?
11. In what mathematical sense is the Ritz approximation using two basis functions better than the one using a single basis function?